

Verified-Performance Air Purifier

Innovation dossier - 1. The idea

PROPOSITION

CONCENTRATE applied to the real 'Perceived value / does it actually clean the air' friction: Optimize for hours-actually-run under real noise tolerance, not peak lab CADR. Assumption challenged: "The device sits in one fixed location, chosen once." -> What if the device (or its effect) followed the person, not the room?

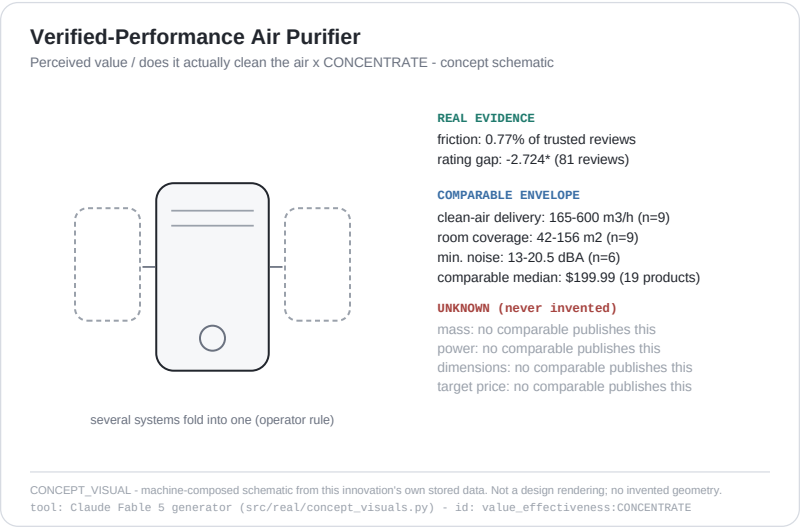


Figure: **CONCEPT_VISUAL**

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

artifact visual-value_effectiveness:CONCENTRATE - /concept-visuals/value_effectiveness_CONCENTRATE.svg

ENGINEERING ENVELOPE

comparable basis: 10 verified official Versuni/Philips air-purifier families (room-scale)

metric	value from verified comparables	epistemic type
performance cadr m3h	165-600 m3/h (n=9 comparables)	OBSERVED_COMPARABLE
room coverage m2	42-156 m2 (n=9 comparables)	OBSERVED_COMPARABLE
acoustic min dba	13-20.5 dBA (n=6 comparables)	OBSERVED_COMPARABLE
target mass kg	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target power w	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
reference market price usd	median \$199.99 (range \$15.95-\$618.0, n=19) - market price of affected products today, not a proposed price	REFERENCE_MARKET_PRICE

State: **CHALLENGED**

Critic verdict CHALLENGE - a real dimension pushes back.

WHY NOW - THE OBSERVED REALITY

0.77% of trusted reviews (81 reviews across 36 real products) carry this friction, with a -2.724* average rating gap. Signal: Whether it 'actually cleans the air' is backed by four real peer-reviewed effectiveness trials.

TARGET USER / CONTEXT

Households whose real reviews carry this friction: 81 reviews across 36 real products (90.1% verified purchase, 2010-06-21-2023-04-24).

UNKNOWN - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

MECHANISM - METHOD CHOICE

CONCENTRATE - Multiple systems -> one system.

METHOD_CHOICE - authored design vocabulary

PRODUCT ARCHETYPE - DESIGN RULE

mobility: Single-unit - operator CONCENTRATE turns several systems into one.

room scale, size class, power class, noise ambition, maintenance burden, filter service dependency, connected: UNKNOWN - no real data in this pipeline for this field.

DESIGN_RULE - each populated field follows from the operator's own fixed definition, never from evidence

THE DERIVATION - THREE PARTS, EACH LABELLED

1. Reality OBSERVED

0.77% of trusted reviews (81 reviews across 36 real products) carry this friction, with a -2.724* average rating gap. Signal: Whether it 'actually cleans the air' is backed by four real peer-reviewed effectiveness trials.

2. Transformation METHOD_CHOICE

METHOD CHOICE, not evidence: the authored operator table assigns CONCENTRATE - "Multiple systems -> one system." - to this friction. Analyst design judgment, applied deterministically.

3. Product consequence RESEARCH_TENSION

Optimize for hours-actually-run under real noise tolerance, not peak lab CADR. Assumption challenged: "The device sits in one fixed location, chosen once." -> What if the device (or its effect) followed the person, not the room?

PARENT PATHS

tension:T1 - Filtration vs. noise-driven adherence TENSION

test: Falsified when one design point achieves RP-01's measured primary-room particulate reduction at a sound level that removes noise as a named reason for running the device less (RP-05) - both halves are stored, quantified findings.

tension:T2 - Constant vs. automatic operation TENSION

test: Falsified when an auto-mode arm reaches the lower bound of RP-07's constant-mode reduction interval while keeping auto's runtime/noise advantage - the two stored confidence intervals would then overlap and the trade-off dissolves.

tension:T3 - CADR theory vs. real-home performance TENSION

test: Falsified when a CADR-derived sizing prediction matches a source-strength-adjusted measured loss rate within the fan-speed band RP-06 reports, in a home with the central air-handling configuration RP-01 identifies as decisive.

assumption:A2 - The device sits in one fixed location, chosen once.

ASSUMPTION_TO_TEST

test: Observe placement behaviour directly: in a small real-home panel, record whether households relocate the unit between rooms within 30 days (RP-01 already shows the benefit leaks unevenly beyond the placed room; RP-04 shows a moving spatial source). The assumption weakens if a material share relocate.

assumption:A5 - Clean-air delivery rate (CADR) is the primary performance basis.

ASSUMPTION_TO_TEST

test: Challenged when a certification body in the trend corpus adopts a loss-rate or realized-exposure metric alongside CADR - concretely, when TC-R03 (AHAM Verifide, the one record whose theme tags contain 'cadr': ['certification', 'cadr', 'iaq_standards']) stops centring on it, or TC-R02 (ENERGY STAR, themes ['regulation', 'energy_efficiency', 'iaq_standards'], CADR present only in its prose scope note) adds a non-CADR performance theme tag. Fragile baseline: both documents carry published_date null, so change detection has no dated anchor.

assumption:A6 - Continuous, constant operation is the default expectation.

ASSUMPTION_TO_TEST

test: Challenged when connected+adaptive share exceeds 25% of the corpus (currently 23/237 = 9.7%), or when an independent trial reproduces RP-07's constant-mode filtration under auto operation.

evidence ids: taxonomy:value_effectiveness, RP-01, RP-02, RP-06, RP-07, TC-R03, TC-R02

FIELD EVIDENCE BEHIND THESE PATHS (TITLES)

- Self-reported health impacts of do-it-yourself air cleaner use in a smoke-impacted community
- Real-World Effectiveness of Portable Air Cleaners in Reducing Home Particulate Matter Concentrations
- Reduction of personal PM2.5 exposure via indoor air filtration systems in Detroit: an intervention study
- The impact of portable air cleaners on indoor particulate matter concentrations and perceptions of indoor air quality: a randomized crossover trial in three multifamily buildings
- Analysis of sources and sinks of indoor particulate matter reveals insights into the real-world efficacy of portable air cleaners in a randomized intervention trial
- PM10, PM2.5, PM1, and PM0.1 resuspension due to human walking

DESIGN DNA

F consumer friction (taxonomy:value_effectiveness) PRESENT

Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator METHOD_CHOICE

Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (value_effectiveness) PRESENT

Whether it 'actually cleans the air' is backed by four real peer-reviewed effectiveness trials.

T scientific tension PRESENT

Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance

R rival gap MISSING_UNVERIFIED

No real competitor-weakness data clears the white-space threshold for this theme.

C versuni capability MISSING_UNVERIFIED

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption PRESENT

The device sits in one fixed location, chosen once.; Clean-air delivery rate (CADR) is the primary performance basis.; Continuous, constant operation is the default expectation.

E economic condition PRESENT

Real price-weighted exposure computed for this theme (\$10,648.05).

ECONOMICS - RELATIVE INDICATORS ONLY

price-weighted exposure: \$10648.05

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$199.99 (n=19 products) - not this concept's price

MEDIAN real listed price across the 19 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

category assumptions: A2, A5, A6

UNCERTAINTIES

- No real Versuni data on how a certification-based trust score would perform against the real complaint text driving this theme's prevalence - the rating is about integration effort, not measured consumer response to the claim.

CONTRADICTIONS

- none recorded for this innovation

OBSERVED VS ESTIMATED VS UNKNOWN

observed: friction stats; envelope: performance cadr m3h, room coverage m2, acoustic min dba; DNA F/S/T/A/E

method choice: operator, archetype design rule; DNA O

unknown: envelope: target mass kg, target power w, target dimensions; persona; DNA R/C

Where this idea stands: **CHALLENGED**

Critic verdict CHALLENGE - a real dimension pushes back.

PROTOTYPE STATE

CONCEPT_VISUAL

CONCEPT_VISUAL means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

NEXT EXPERIMENT

Evidence that AHAM/ENERGY STAR certification specifically addresses this theme's real complaint pattern (not just CADR broadly) would strengthen this rating further.

KILL CRITERION

Formal-case kill criterion applies (see formal_case.abandon_signal)

CRITIC DIMENSIONS

EVIDENCE	CHALLENGE	Gate passed but prevalence (0.77%) is only marginally above the materiality floor - thin evidence base.
ECONOMIC	CHALLENGE	Real price-weighted exposure (\$10,648) is below the corpus median (\$16,113) - a real but comparatively modest signal.
COMPETITIVE	NEEDS_EVIDENCE	No real rival-weakness data recorded for this friction theme.
HUMAN	NEEDS_EVIDENCE	No real usability/human-factors evidence exists in this pipeline for this concept.
PHYSICAL	NEEDS_EVIDENCE	No real lab/engineering feasibility data exists in this pipeline for this concept.
VERSUNI_FIT	NEEDS_EVIDENCE	No real organizational-capability assessment exists in this pipeline for this concept.
TIMING	NEEDS_EVIDENCE	No real market-timing/window evidence exists in this pipeline for this concept.
ROBUSTNESS	NEEDS_EVIDENCE	No real sensitivity/robustness study exists in this pipeline for this concept.

critic overall: CHALLENGE

CRITERIA RESULTS

NEEDS_EVIDENCE: 33 - PASS: 14 - N/A: 3 - KILL: 1 - CHALLENGE: 1

C1	KILL	Not white space.
B5	CHALLENGE	Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance

LINEAGE / PROVENANCE

generated by: src/real/innovations_real.py
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figure source: /concept-visuals/value_effectiveness_CONCENTRATE.svg (generated 2026-08-31T12:16:37.702869+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)
data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)
dossier generator: src/real/generate_innovation_dossiers.py